



CSE 3313: Computer Architecture

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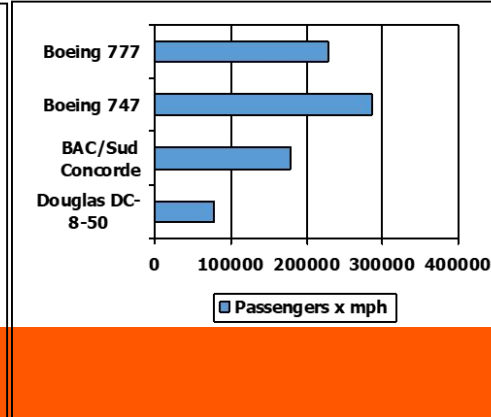
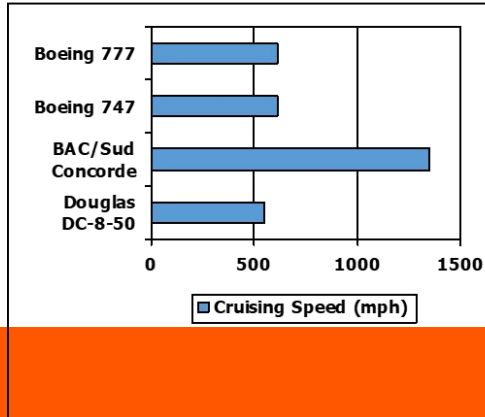
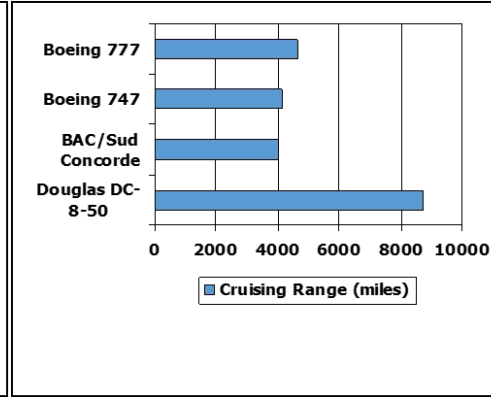
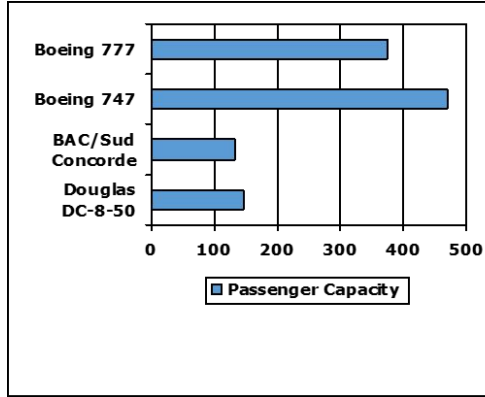
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Criteria for Performance



Airplane	Capacity [persons]	Range [km]	Cruising speed [km/h]	Throughput [pers·km/h]
Boeing 737	240	4828	907	217680
BAC/Sud Concorde	132	6437	2172	286704
Boeing 777-200LR	301	15120	892	268492
Airbus A380-800	853	13642	944	805232

Which Airplane has the best performance



Criteria for Computer Performance



- **Processor speed:** The speed of the processor, measured in GHz (gigahertz), determines how quickly the computer can execute instructions and process data.
- **Memory:** The amount and speed of the memory, including RAM (random access memory) and cache memory
- **Storage:** The speed and capacity can impact the speed at which data can be stored and retrieved.
- **I/O devices:** The speed and efficiency of input/output devices,
- **Software optimization:** The efficiency of the software running on the system, including operating systems and applications,

Response Time

Response time is the time from the start to completion of a task. This also includes:

- Operating system overhead.

- Waiting for I/O and other processes

- Accessing disk and memory

- Time spent executing on the CPU or execution time.



Throughput



Throughput is the total amount of work done in a given time.

- CPU execution time is the total time a CPU spends computing on a given task.
- It also excludes time for I/O or running other programs. This is also referred to as simply CPU time.

Throughput vs Response Time



Performance is determined by execution time as performance is inversely proportional to execution time.

Performance = (1 / Execution time) “X is n time faster than Y”
[[Performance_X]] / [[Performance_Y]]
= [[Execution time_Y]] / [[Execution time_X]] = n

Relative Performance



Example: time taken to run a program

10s on A, 15s on B

$$P_a/P_b = E_b/E_a$$

Execution Time_A / Execution Time_B

$$= 15s / 10s = 1.5$$

So A is 1.5 times faster than B

Defining Execution Time

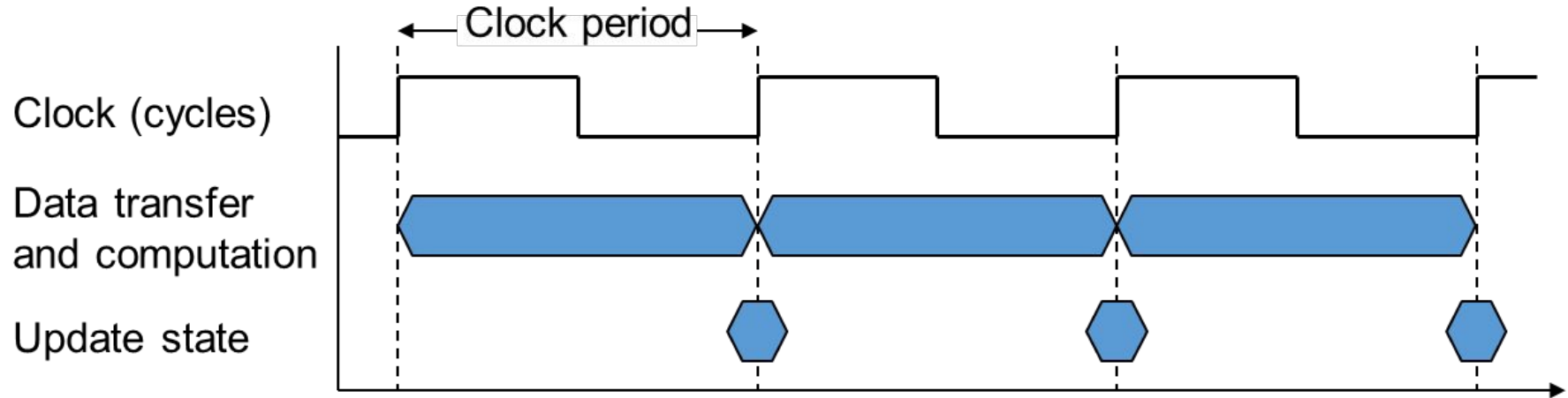
Elapsed time

- Total response time, including all aspects
- Processing, I/O, OS overhead, idle time
- Determines system performance

CPU time

- Time spent processing a given job
- Discounts I/O time, other jobs' shares
- Comprises user CPU time and system CPU time
- Different programs are affected differently by CPU and system performance

Clock Period and Clock Rate



Clock Period and Clock Rate



Clock period: duration of a clock cycle

e.g., $250\text{ps} = 0.25\text{ns} = 250 \times 10^{-12}\text{s}$

Clock frequency (rate): cycles per second

e.g., $4.0\text{GHz} = 4000\text{MHz} = 4.0 \times 10^9\text{Hz}$

You need to remember this for Mid Term!



Symbol	Number	Exponent
1 Hz (hertz)	1 Hz	1 Hz
1 kHz (kilohertz)	1000 Hz	$1 \cdot 10^3$ Hz
1 MHz (megahertz)	1,000,000 Hz	$1 \cdot 10^6$ Hz
1 GHz (gigahertz)	1,000,000,000 Hz	$1 \cdot 10^9$ Hz

Clock Period and Clock Rate Math



Program A runs on Computer C

The clock frequency of Computer C is 10 Hz.

The Clock Pulse Requirement for each instruction in program A is given below:

Add- 4 Clock Pulse

Sub- 2 Clock Pulse

Multiplication - 6 Clock Pulse

Program A
Add
Add
Sub
Sub
multiplication

Clock Period and Clock Rate Math



Program A runs on Computer C

The clock frequency of Computer C is 10 Hz.

Add- 4 Clock Pulse

Sub- 2 Clock Pulse

Multiplication - 6 Clock Pulse

Calculate total clock pulse:

$$4 + 4 + 2 + 2 + 6 = 18$$

Program A
Add
Add
Sub
Sub
multiplication

Clock Period and Clock Rate Math



Program A runs on Computer C

The clock frequency of Computer C is 10 Hz.

Add- 4 Clock Pulse

Sub- 2 Clock Pulse

Multiplication - 6 Clock Pulse

Calculate the time period for each clock cycle

$T = 1/f$, $f = 10$, $T = 1/10 = 0.1s$

Program A
Add
Add
Sub
Sub
multiplication

Clock Period and Clock Rate Math



Time for each clock pulse = 0.1 s

Total pulse = 18

Time required to run the program = $18 \times .1 = 1.8$ s

If we formulate it, **Time required to run a program in a computer = Total clock pulse for the program x Time required for each clock pulse**

Or, total clock pulse for the program / frequency

Program A
Add
Add
Sub
Sub
multiplication

CPU Time

$$\begin{aligned}\text{CPU Time} &= \text{CPU Clock Cycles} \times \text{Clock Cycle Time} \\ &= \frac{\text{CPU Clock Cycles}}{\text{Clock Rate}}\end{aligned}$$



Performance improved by

- Reducing number of clock cycles
- Increasing clock rate
- Hardware designer must often trade off clock rate against cycle count

- | Instruction | add | sub | addi |
|-------------------------------|-----|-----|------|
| CPI for instruction
Cycles | 1 | 2 | 3 |
| IC in sequence 1 | 2 | 1 | 2 |
| IC in sequence 2 | 4 | 1 | 1 |

A Practice Problem



Our favorite program runs in 10 seconds on computer A, which has a 400 MHz clock. We are trying to help a computer designer build a machine, B, that will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for this program. What clock rate should we tell the designer to target?

A Practice Problem (Cont.)



Our favorite program runs in 10 seconds on computer A, which has a 400 MHz clock. We are trying to help a computer designer build a machine, B, that will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for this program. What clock rate should we tell the designer to target?

$$\begin{aligned}\text{CPU time}_A &= \frac{\text{CPU clock cycle}_A}{\text{Clock rate}_A} \\ 10 \text{ Seconds} &= \frac{\text{CPU clock cycle}_A}{400 \times 10^6 \text{ cycles/sec}}\end{aligned}$$

A Practice Problem (Cont.)



$$\text{CPU time}_A = \frac{\text{CPU clock cycle}_A}{\text{Clock rate}_A}$$

$$10 \text{ Seconds} = \frac{\text{CPU clock cycle}_A}{400 \times 10^6 \text{ cycles/sec}}$$

$$\begin{aligned}\text{CPU clock cycle}_A &= 10 \text{ seconds} \times 400 \times 10^6 \text{ cycles/sec} \\ &= 4000 \times 10^6 \text{ cycles}\end{aligned}$$

A Practice Problem (Cont.)



Our favorite **program runs in 10 seconds on computer A**, which has **a 400 MHz clock**. **We are trying to help a computer designer build a machine, B, that will run this program in 6 seconds**. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for this program. What clock rate should we tell the designer to target?

$\text{CPU time}_B = 6 \text{ Seconds}$

A Practice Problem (Cont.)



Our favorite **program runs in 10 seconds on computer A**, which has a **400 MHz clock**. **We are trying to help a computer designer build a machine, B, that will run this program in 6 seconds**. The designer has determined that a substantial increase in the clock rate is possible, **but this increase will affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for this program**. What clock rate should we tell the designer to target?

$$\text{CPU clock cycle}_B = 1.2 \times \text{CPU clock cycle}_A$$

A Practice Problem (Cont.)



$$\begin{aligned} 6 \text{ seconds} &= \frac{1.2 \times 4000 \times 10^6 \text{ cycles}}{\text{Clock rate}_B} \\ \text{Clock rate}_B &= \frac{1.2 \times 4000 \times 10^6 \text{ cycles}}{6 \text{ seconds}} \\ &= 800 \text{ MHz} \end{aligned}$$

Machine B must therefore have twice the clock rate of A to run the program in 6 seconds.

CPI = Clock Per Instruction

- If different instruction classes take different numbers of cycles

$$\text{Clock Cycles} = \sum_{i=1}^n (\text{CPI}_i \times \text{Instruction Count}_i)$$

■ Weighted average CPI

$$\text{CPI} = \frac{\text{Clock Cycles}}{\text{Instruction Count}} = \sum_{i=1}^n \left(\text{CPI}_i \times \underbrace{\frac{\text{Instruction Count}_i}{\text{Instruction Count}}}_{\text{Relative frequency}} \right)$$

CPI Example Problem

Computer A: Cycle Time = 250ps, CPI = 2.0

Computer B: Cycle Time = 500ps, CPI = 1.2

Same ISA, Which is faster, and by how much?

$$\begin{aligned}\text{CPU Time}_A &= \text{Instruction Count} \times \text{CPI}_A \times \text{Cycle Time}_A \\ &= 1 \times 2.0 \times 250\text{ps} = 1 \times 500\text{ps}\end{aligned}$$

$$\begin{aligned}\text{CPU Time}_B &= \text{Instruction Count} \times \text{CPI}_B \times \text{Cycle Time}_B \\ &= 1 \times 1.2 \times 500\text{ps} = 1 \times 600\text{ps}\end{aligned}$$

$$\frac{\text{CPU Time}_B}{\text{CPU Time}_A} = \frac{1 \times 600\text{ps}}{1 \times 500\text{ps}} = 1.2$$



Thank you!